METRIC

MIL-C-14200C
10 November 1988
SUPERSEDING
MIL-C-14200B
29 September 1978

MILITARY SPECIFICATION

CONTAINERS, SHIPPING AND STORAGE: METAL, REUSABLE; FOR ENGINES, TRANSMISSIONS, DIFFERENTIALS, TRANSFERS, AND SIMILAR ASSEMBLIES (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of metal, reusable containers (boxes) used for shipment and storage of engines, transmissions, differentials, transfers, and other similar automotive components (see 6.1).

1.2 Classification. Containers covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Pressurized container.
Type II - Controlled breather container.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8145

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1261	- Arc Welding Procedures for Constructional Steels.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

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3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free of defects and imperfections that might affect the serviceability and function of the finished product (see 4.8.1).

3.2.1 Recycled, virgin, and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.8.1 and 6.4.3).

3.3 Design and construction. All containers furnished under this specification shall be fabricated in accordance with the applicable drawing (see 4.8.1, 4.8.2, and 6.2).

3.3.1 Welding. Unless otherwise specified (see 6.2), welding shall conform to MIL-STD-1261. All fluxes, acids, and flux solvents shall be removed prior to preparation for painting. Peening of welds in any pressurized area shall not be permitted. Defective areas of weldments shall be removed by any applicable means and then rewelded (see 4.8.1 and 4.8.2).

3.4 Performance.

3.4.1 Air leakage.

3.4.1.1 Type I. Containers shall show no leakage when pressurized to 69 kilopascals (kPa) air pressure (gage) (see 4.8.3.1).

3.4.1.2 Type II. Containers shall show no leakage when pressurized to 34.5 kPa air pressure (gage) (see 4.8.3.2).

3.4.2 Rough handling. Containers shall be fabricated in such a way that contents shall not be subject to damage. Subsequent to rough handling tests, containers shall meet the requirements of 3.4.1.1 and 3.4.1.2, as appropriate (see 4.8.4).

3.5 Painting. Unless otherwise specified (see 6.2), preparation and painting shall be in accordance with specifications referenced on the applicable drawing. The interior and exterior topcoat shall conform to MIL-C-46168 or MIL-C-53039 (see 4.8.2).

3.6 Marking. Identification and marking shall be in accordance with the applicable drawing (see 4.8.2).

3.7 Workmanship. Workmanship shall be of quality necessary to produce containers free of defects that will affect their strength, assembly, appearance, serviceability, and durability, as well as freedom from burrs, sharp edges, or any condition that might present a safety hazard to operating or maintenance personnel (see 4.8.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (QCI) (see 4.5).
c. Control tests (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 25 $\pm$ 10 degrees Celsius (^{o}C).
b. Atmospheric pressure: Site pressure.
c. Relative humidity: Uncontrolled room ambient.

4.4 First article inspection. When first article is required (see 3.1), first article inspection shall be performed on first article samples (see 6.2 and 6.3). Unless otherwise specified (see 6.2), first article inspection shall include the inspections specified in table I. The air leakage test (see 4.8.3) shall be performed after the rough handling tests (see 4.8.4).

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TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	QCI		Control tests	Inspection of packaging
				Examination	Test		Examination
Materials and construction	3.2 thru 3.3.1	4.8.1	X				
Examination	3.3, 3.3.1, 3.5, 3.6 and 3.7	4.8.2	X	X			
Air leakage	3.4.1	4.8.3	X		X	X	
Rough handling	3.4.2	4.8.4	X			X	
Packaging examination	5.1 and 5.3	4.7.1	X				X

4.5 QCI. QCI shall be conducted on the sample items and shall include the examination of 4.5.1 and the tests of 4.5.2 (see 6.2). Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.5.1 Examination. The sample items shall be examined for the characteristics specified in table II (see 6.2).

TABLE II. Examination of characteristics.

Characteristic	Requirement
Dimensions not as specified or out of tolerance, interchangeability affected.	3.3
Dimensions not as specified or out of tolerance, interchangeability not affected.	3.3
Welding not as specified.	3.3.1
Painting not as specified.	3.5
Marking not as specified.	3.6
Workmanship not as specified, serviceability affected.	3.7
Workmanship not as specified, appearance affected.	3.7

4.5.2 QCI tests. Subsequent to the examination of 4.5.1, the sample items shall be subjected to the QCI test specified in table I.

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4.6 Control tests. Sample items selected for control tests shall be new and examined in accordance with 4.5.1 prior to subjecting the sample items to the control tests specified in table I (see 6.2). Tests shall be performed in the following order: 4.8.3, 4.8.4.1, 4.8.4.2, 4.8.4.3, 4.8.4.4, 4.8.4.5, 4.8.3, and 4.8.4.6.

4.7 Inspection of packaging.

4.7.1 Packaging examination. The sample items shall be examined for the packaging characteristics specified in table III (see 6.2).

TABLE III. Examination of packaging characteristics.

Packaging characteristic	Requirement
Securement of mounts not as specified.	5.1
Securement and location of accessories not specified.	5.1
Torquing of bolts not as specified.	5.1
Coating not as specified.	5.1
Pressure not as specified.	5.1.1 and 5.1.2
Marking - omitted, incomplete, incorrect, illegible, improper size, location sequence, and method of application.	5.3

4.7.2 Government inspection. The Government, at unscheduled intervals, may inspect all materials and processes to determine conformance to the requirements of section 5 and documents referenced therein. Any evidence of deviation from the specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3.1 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Examination. Conformance to 3.3, 3.3.1, 3.5, 3.6, and 3.7 shall be determined by the examination specified in table II.

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4.8.3 Air leakage.

4.8.3.1 Type I. To determine conformance to 3.4.1.1, each container shall be assembled and closed. The relief valve shall be blocked off. The container shall be filled to 69 kPa air pressure (gage) and placed in water. The container shall be rotated until all surfaces have been submerged and examined.

4.8.3.2 Type II. To determine conformance to 3.4.1.2, each container shall be assembled and closed. A pressure valve shall be substituted for the breather valve. The container shall be filled to 34.5 kPa air pressure (gage) and placed in water. The container shall be rotated until all surfaces have been submerged and examined.

4.8.4 Rough handling. To determine conformance to 3.4.2, containers shall be tested as specified in 4.8.4.1 through 4.8.4.6. Unless otherwise specified (see 6.2), containers selected for testing shall be packed with components for which the containers were designed. Unless otherwise specified (see 6.2), the contractor shall provide the component. The container shall be sealed. When specified (see 6.2), for very expensive and security items, a container packed with a "dummy load" shall be tested prior to testing a container packed with a "live load" (see 6.4.1 and 6.4.2).

4.8.4.1 Edgewise drop test. The packed container shall be supported at one end of its base on a wood sill or block, 6 inches nominal [15 centimeters (cm)] in height, and placed at right angles to the skids. The opposite end of the container shall be raised and allowed to drop freely from heights of 30, 60, and 90 cm successively onto a concrete or metal surface. The test shall be applied to each end of the container. If the size of the container and the location of the center of gravity are such that drop tests cannot be made from all of the prescribed heights, the greatest attainable height shall be the height for succeeding drops until a total of three drops have been accomplished.

4.8.4.2 Cornerwise drop test. The packed container shall be supported at one corner of its base on a block, 6 inches nominal (15 cm) in height. A block, 12 inches nominal (30 cm) in height, shall be placed under the other corner of the same end of the container. The lowest point of the opposite end of the container shall then be raised and allowed to fall freely from heights of 30, 60, and 90 cm successively onto a concrete or metal surface. If the size of the container and the center of gravity are such that drop tests cannot be performed from all the prescribed heights, the greatest attainable height shall be the height for succeeding drops until a total of three drops have been performed. This test shall be applied on two diagonal corners at the bottom of the container.

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4.8.4.3 Tip over test. The packed container, erect on its base on a hard level floor, shall be slowly tipped to the side until it falls freely and solely by its own weight to the floor. After righting the container, the test shall be repeated on the opposite side. Cylindrical containers shall be tipped at right angles to the longitudinal axis of the installed item and allowed to fall freely to the floor.

4.8.4.4 Impact test. An impact test shall be applied to each end of the packed container. The container shall be suspended as a pendulum at the end of four or more cables (two cables shall be used for cylindrical containers with two lifting eyes). The cables shall be of sufficient length to prevent any interference or binding. A flat, vertical, stationary masonary or metal barrier, with a thickness of not more than 5 cm of wood between the barrier and container, shall be provided for the container to strike against. The suspended container shall be raised vertically to a height which will allow the lowest point of the container (while swinging through the arc of the pendulum) to clear the floor. While the suspended container is resting lightly against the barrier and prior to pulling back for the impact, the center of balance shall be marked (if not stenciled on the container) as a measuring reference point. This mark shall be placed at the lowest point on the container shell. The suspended container shall be pulled back with a straight even pull until a height of 46 cm plus the aforementioned floor clearance is reached. This measurement shall be taken from the measuring reference point on the container to the floor. At this point the container shall be released in a manner to allow a smooth even travel to the barrier.

4.8.4.5 Flatwise drop test. The packed container shall be raised so that its base is parallel to the floor and allowed to fall freely once from a height of 15 cm and once from a height of 30 cm to a concrete or metal surface.

4.8.4.6 Examination and functional test. Containers tested as specified in 4.8.4.1 through 4.8.4.5 shall be opened and the contents examined for visible signs of movement and damage. When specified (see 6.2), a functional test shall be conducted on the contents if the contents are a "live load".

5. PACKAGING

5.1 Preservation. Each container shall be completely assembled for shipment (see 5.1.1 and 5.1.2). Item mounting plates or brackets shall be secured in place. Any other accessories shall be securely fixed at any convenient place on the interior of the container. Closure bolts or other closure devices shall be assembled and drawn tight in accordance with the torque requirements of the applicable drawing. All exposed operating surfaces of closure devices and exposed threaded surfaces shall be coated with a compound conforming to grade 4 of MIL-C-16173 (see 4.7.1).

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5.1.1 Type I. Containers shall be pressurized to 34.5 kPa (gage) (see 4.7.1).

5.1.2 Type II. Containers shall not be pressurized (see 4.7.1).

5.2 Packing. Application or use of exterior container is not required.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), the container shall be marked in accordance with the requirements of MIL-STD-129 (see 4.7.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Containers conforming to this specification are intended for use as humidity controlled containers for shipment and storage of tank and automotive engines, transmissions, differentials, transfers, and similar assemblies.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type I or type II containers (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. If first article is required (see 3.1).
e. Title, number, and date of applicable drawing (see 3.3).
f. Welding, if other than as specified (see 3.3.1).
g. Preparation and painting, if other than as specified (see 3.5).
h. Responsibility for inspection, if other than as specified (see 4.1).
i. Place of inspection, if other than as specified (see 4.1).
j. First article inspection categories (see 6.3), sample size for each category, and specific tests for each sample.
k. First article inspection, if other than as specified (see 4.4).
l. Arrangement for first article inspection, approval of test results, and disposition of first article (see 6.3).
m. Governments rights and conditions for waiving first article inspection (see 6.3).
n. Sample size for QCI examination and tests (see 4.5).
o. QCI acceptance criteria (see 4.5).
p. Control test sample size, frequency of tests, and disposition of lots covered by the sample size and of defective items (see 4.6).

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q. Sample size for packaging examination (see 4.7.1), acceptance criteria, and disposition of defective items.
r. If containers should not be packed with the components (see 4.8.4).
s. If the Government is to supply the components (see 4.8.4).
t. If a container should be packed and tested with a "dummy load" (see 4.8.4).
u. If functional testing of component is required (see 4.8.4.6).
v. Special marking, if required (see 5.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the first article samples should be a first article sample, preproduction sample, initial production sample, first production item, or standard production item from the contractor's current inventory and for each inspection category specify the number of samples to be inspected and the specific tests to be performed on each sample. The contacting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Dummy load. A "dummy load" is a load of the same size, weight, and weight distribution as the component (see 4.8.4).

6.4.2 Live load. A "live load" is a load that contains the component for which the container was designed (see 4.8.4).

6.4.3 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.4).

6.4.4 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or

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industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Subject term (key word) listing.

CARC paint topcoats
Container, controlled breather
Container, pressurized

6.6 Metrication. All units were converted to metric in accordance with FED-STD-376, revision A, dated 5 May 1983.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Preparing activity:
Army - AT

(Project 8145-0033)

Review activities:
Army - SM
Navy - MC
Air Force - 84

User activities:
Army - GL, ME

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